
Group 1: Predicting Customer Conversion in Digital Marketing Campaigns

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Abstract

In order to find the driving factors of customer conversion and to find out whether the agency can boost customer conversion rate, one statistical model and three predictive models were trained on the pre-processed and cleaned dataset. The models used were logistic regression, random forest, support vector machine, and gradient boosting. Logistic regression helped interpret the relationship between the variables and customer conversion and showed that if the agency can find ways to increase the engagement metrics measured from the customers, then the customer conversion rate will increase. Gradient boosting yielded the best performance out of the three predictive models regarding predictive accuracy and is the ideal choice if the agency wishes to predict future customer conversions.

1 Introduction

1.1 Project overview

Digital marketing today is an important part of any marketing campaign. It is a primary source of customers for many companies, and proper research of customer behavior and the effectiveness of different digital marketing campaigns is essential. With the amount of data available online, the data science field had also become a very useful instrument for any company that is looking to optimize its online presence.

The digital marketing agency Market Pulse Analytics has observed declining conversion rates across several recent online campaigns. In order to investigate potential drivers of customer engagement and conversion, as well as to predict whether a targeted customer will ultimately convert in future campaigns, the management commissioned the creation of a comprehensive dataset combining demographic information, marketing campaign metrics, customer engagement indicators, and historical purchase behavior. The goal of this project is to analyze the resulting dataset for any correlations or connections that would assist the agency in boosting customer conversion.

The main research question was: can the agency boost customer conversion rate using the metrics that were collected in the dataset?

The objectives of the project were split into four parts:

- **Pre-Processing, Cleaning, and Identifying Potential Biases:** A standard check for any missing values, outliers, duplicates, or any other discrepancies in the dataset.
- **Exploratory Data Analysis and Visualization:** An overview of the cleaned data and search for any linear correlations between the features in the dataset. Visualization of the core findings.
- **Statistical Modelling:** A more in-depth analysis of the dataset features with the main purpose of finding the main drivers of customer conversion and non-linear relationships between the data.

- Predictive Modelling: Creation of three different machine learning classification models and their comparison. Analysis of the results, hyperparameter tuning, and the selection of the best model for the agency's goals.

1.2 Background and related literature

Since the broad purpose of this project is to find any relevant information about conversion drivers in the agency's dataset, it is beneficial to first understand what conversion is. Conversion is a company turning a visitor into a customer[1, Key Terms]. The second most important thing is to understand how the process of conversion works.

Before a visitor becomes a customer, several stages must take place. Philip Kotler and Gary Armstrong[2, p. 169] describe these stages as:

- Need recognition.
- Information search.
- Evaluation of alternatives.
- The purchase decision.
- Postpurchase behavior.

The Buyer Decision Process begins when a consumer recognizes a problem. Once the problem is identified, the consumer might begin searching for a solution. This is when the consumer interacts with the company's digital marketing for the first time. Although personal recommendations from friends and family members usually have much greater influence on the consumer's decision.

In the next stage, the consumer evaluates the available alternatives. They compare such metrics as, for example, price, quality, customer service, popularity, etc. The evaluation process might differ significantly between consumers. When this process is completed, the consumer decides whether they want to purchase the offered product or not.

If the consumer decides to go through with the purchase and becomes a customer, they assess whether the product met all of their expectations. Customer's satisfaction after the purchase is also very important to the company, because loyal customers might buy the product again or recommend it to their friends. On the other hand, dissatisfied customers are very likely to advise their friends against buying the products or services of that specific company.

This aligns with the agency's different campaign types: Awareness, Consideration, Conversion, and Retention.

1.3 Possible approaches

Before beginning a new research, we must understand which existing approaches have already been used for similar problems. Existing marketing research uses diverse methods to determine the drivers of customers' conversion and evaluate the success of digital marketing campaigns.

The initial step in conducting a marketing study is performing an exploratory data analysis (EDA) to detect trends or linear relationships present in the data. Descriptive statistics and visualization methods are often used to show how customer characteristics, engagement metrics, and campaign characteristics relate to customers' conversion rates.

In addition to exploratory analysis, statistical-model building methods have also been used to understand the relationship between customers' conversion rates and various variables. Logistic regression models are widely used in marketing because they are applicable for modelling the likelihood of a binary outcome, such as whether a customer converted or did not convert. Logistic regression allows researchers to both estimate each customers' probability of converting, and evaluate the impact each feature has on the likelihood of conversion.

Recently, machine-learning techniques have become increasingly popular in marketing. Classification algorithms such as decision trees, random forests, and support vector machines are regularly used by researchers to predict customers' responses to marketing campaigns. These algorithms have the ability to identify complex/dynamic relationships among independent variables in large data sets and often achieve greater accuracy than traditional statistical models. As a result, there is a trend among

researchers to combine statistical analyzes and machine-learning techniques to conduct studies of the customer conversion process in digital marketing contexts.[3, Chapter 3]

2 Preprocessing

2.1 Missing Values

After importing the dataset, we first checked for missing values. Our examination revealed that all 79 missing values were in the "Income" column, it represents approximately 1% of the total dataset, and no missing values were found in the other columns. To further reveal any potential relationships for NA values within the overall variable, we created an additional missing values index for the income variable and compared it with categorical and numerical variables. The results showed that missing values for the *Income* variable were relatively evenly distributed across categories such as *Gender*, *CampaignChannel*, *CampaignType*, and *ConversionRate*. Only the *Age* variable showed a small but statistically significant difference in missing values, while there was no significant correlation between the *AdSpend* and *CampaignChannel* variables and NA values in the *Income* variable. Overall, the number of NA values in the *Income* variable was small and unlikely to have a substantial impact on the analysis results.

2.2 Duplicate Entries

A duplicate entry check was then performed on the dataset. This revealed 200 duplicated CustomerID values. To ensure these records were complete duplicates (not a simple data entry error), further checks were conducted, revealing that these 200 duplicate Customer IDs corresponded exactly to 200 completely duplicate rows in the dataset. Since these rows were exact duplicate records with no additional information, they were removed from the dataset. After removing duplicate rows, the dataset contained 8000 observations and 19 variables.

2.3 Validity and Consistency Checks

Next, we performed validity and consistency checks on the numeric variables. For the income variable, we used the interquartile range (IQR) method to identify potential outliers. This method detected two extreme values, both $1e+21$. Since this is not a reasonable annual income value, we considered these values invalid and recoded them as NA. Furthermore, we examined the summary statistics of key numerical variables, including *Age*, *Income*, *AdSpend*, *ClickThroughRate*, *ConversionRate*, *WebsiteVisits*, *PagesPerVisit*, *TimeOnSite*, *SocialShares*, *EmailOpens*, *EmailClicks*, *PreviousPurchases*, and *LoyaltyPoints*. The results showed that the *ConversionRate* contained two values outside the valid range of 0 to 1. Therefore, we considered these values erroneous and recoded them as NA.

2.4 Potential Biases and Representativeness

We have examined missing values, duplicates, and implausible/out-of-range values as internal data quality controls. However, these primary checks help assess internal quality, but do not by themselves establish representativeness. Since we only have the final data tables and are unaware of factors such as sampling design and data collection protocol, we cannot fully verify whether the sample represents a broader target customer group and all future marketing campaigns. The results are useful for the currently provided data sample, but caution is advised when extending these findings to all customers or future campaigns.

2.5 Outcome of Preprocessing

Overall, the preprocessing stage addressed missing values, duplicate entries, and validity issues in the dataset. Missing values were limited to the *Income* and had a minor impact. Completely duplicated rows were removed, and invalid entries (including implausible *Income* values and out-of-range *ConversionRate* values) were recoded as NA. Therefore, the cleaned dataset was considered suitable for the subsequent stages of the project, namely exploratory analysis and modelling.

3 Strategy and Methodology

3.1 Exploratory Data Analysis and Visualization

We began the analysis by exploring the data on a surface level. The first thing that looked odd was the fact that 87% of the customers in the dataset converted. According to the analysis done by WordStream, the average conversion rate across 16000 different US marketing campaigns in 2025 was around 7.5%[4]. This gap might mean that the agency is not calculating conversion in the usual way, and the whole analysis might be affected by this limitation.

Most of the data follows a uniform distribution, including the different engagement metrics and the distribution of campaign types and channels. This is beneficial for the analysis since it eliminates potential biases.

We also found a linear correlation between the amount of budget spent on the campaign and customer conversion.

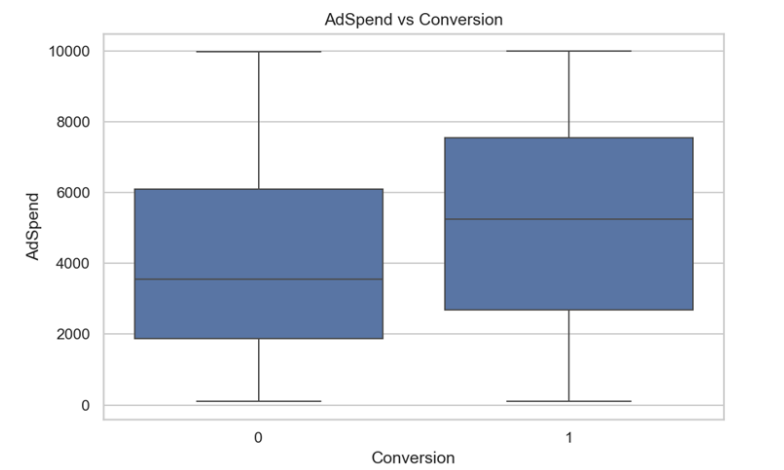


Figure 1: Ad Spend vs Conversion

There was a similar correlation between engagement metrics and customer conversion, meaning that if a customer interacts with marketing content more, there is a higher chance that they will convert.

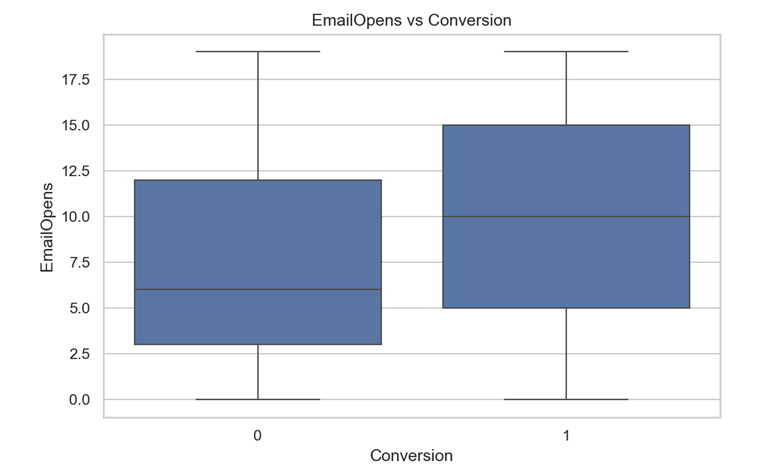


Figure 2: Email Opens vs Conversion

We could not find any other linear correlations within the dataset, but there could still be some non-linear correlations that predictive models might identify.

After attempting to analyze the dataset more thoroughly, we realized that it lacks information about the agency's actions. We established that there was a correlation between engagement metrics and the customer conversion rate, but we could not determine the reason why customers might want to interact with marketing campaigns more since the agency did not provide us with any specifics about the campaigns or how they are managed. The dataset only has information about ad spend, campaign types, and channels, but this is not enough for a comprehensive analysis of the agency's actions. Nevertheless, after examining these features, we found that the Conversion campaign type performs better than the other types. Further analysis showed that there is a difference in how different campaign types react to the increase in ad spend. While most campaign types strongly benefit from ad spend being higher than 5000, the conversion campaign type performs very well even with low ad spend and does not seem to get any major benefit from its increase.

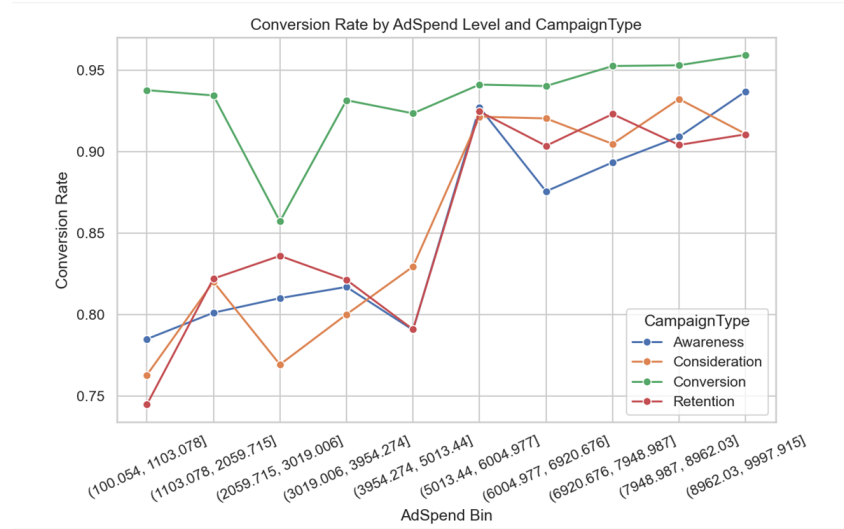


Figure 3: Campaign Types performance in relation to the amount of ad spend

3.2 Statistical Modelling

Logistic regression is the chosen model for statistical modelling for the task at hand. Logistic regression excels at modelling and explaining relationships between other variables and the target variable via log odds. It can also determine whether or not the relationship is statistically significant.[5] Logistic regression was chosen over something like linear regression since logistic regression is efficient and easy to implement and train for binary classification cases and the results are also easy to interpret. For this case of predicting customer conversion, a train test split of 0.8 and 0.2 was created from the preprocessed data, and the logit model was trained and tested on these splits. The performance of the model was then evaluated with the area under the curve of a ROC curve.

3.3 Predictive Modelling

3.3.1 Random Forest

As our first predictive model, we chose Random Forest Classifier. This model works by using a large number of decision trees to come up with a final prediction. Each decision tree is trained on a different subset of data, but the final prediction of the model is a combination of the results of the different trees. This model works better than the use of a single decision tree since it has a lower risk of overfitting, but at the same time, it can handle complex data.

Another reason why the Random Forest Classifier model was chosen as a baseline model is the fact that the model is very flexible and reliable. This model works well for both numerical and categorical data, and it can capture non-linear relationships. This model does not require the data to come from a particular distribution. This model also works well since it has several hyperparameters that can be adjusted to ensure optimal performance.

The model was trained using 80/20 train test split. Several hyperparameters were tuned, such as `n_estimators`, `max_depth`, `min_samples_split`, `class_weight`, and `threshold`. Hyperparameters that gave a noticeable boost to the model's performance were `max_depth` (at 10), `class_weight` (at balanced and balanced_subsample), and `threshold` (at 0.7).

3.3.2 Support Vector Machine

The second chosen predictive model is support vector machine or svm for short. Svm excels at binary classification tasks with a high dimensional dataset. It is also good at tasks that contains non-linear patterns. Similar to logistic regression, svm was also done with a 0.8 and 0.2 train test split on the preprocessed dataset. The model was trained and tuned on the train split and tested on the test split. There were three parameters that were tuned for svm, they are C, Sigma, and the classification threshold. C is the regularization parameter, which means that a smaller c value results in stronger regularization and a larger c value leads to a weaker regularization. If c is too small, overfitting may occur. Sigma controls how local the influence of each data point is. Lower sigma leads to smoother decision boundaries and larger sigma leads to more squiggly or jagged decision boundaries. The classification threshold is the how confident the model has to be in order to classify the data point as converted.

Both sigma and c was tuned via grid search. The sigma candidates were chosen with the sigest function in R which estimates a reasonable range of kernel widths from the training data. It typically returns a small set of candidate values, often the lower, mid, and upper estimates. C's candidates were 2^n where n ranges from -2 to 6. The model then does a 5 fold cross validation that was repeated 3 times in order to find the most optimal combination of sigma and c. The classification threshold were then tuned based on the youden index of the different combinations of sigma and c. In other words, the classification threshold maximises the area under the curve of the ROC curve for each sigma/c combination. This results in a unique threshold for each tested combination of sigma and c. After the model was trained and tuned, each possible combination of sigma and c from the grid search had a threshold tuned for them and their performance metrics were recorded for further analysis. These only includes combinations with $c \leq 8$ since the model often runs into issues for c and larger than 8. The final tuned hyperparameters are: sigma = 0.0347, c = 1, and threshold = 0.8432

3.3.3 Gradient Boosting

The initial plan was to choose logistic regression as the baseline model as one of the 3 predictive models. However, given that we already used logistic regression for statistical modelling and we already had support vector machines and random forests, gradient boosting was chosen instead as the final comparison model. It belongs to the boosting-based tree model family and is suitable for binary classification. Considering the complexity of our dataset and its ability to handle non-linear relationships and variable interactions, it is well-suited for tasks such as customer conversion prediction. Also for the reason which not going with something more complex and choosing the Xboosting model is due to the trade-off between the commercial nature of the project, model parameter complexity, and model interpretation transparency. Therefore, GBM was ultimately chosen as our third model.

First, we load the dataset cleaned in the preprocessing stage. Then, to maintain consistency in evaluation, we adopted a similar setup as before, remove *CustomerID* and *Income_missing*, convert *Conversion* and categorical predictors to factors, drop_na(), and performing an 80/20 stratified training-test set split. 5-fold cross-validation was used during training, and ROC was used as the tuning metric. After prediction, we converted the probabilities into yes/no classification rules. We used the `caret::train()` function with `tuneLength = 5` to automatically adjust the parameters, best hyperparameters taken from `gbm_model$bestTune`. Thresholds of 0.5, 0.7, and 0.9 were compared, resulting in three different CSV outputs. After obtaining the predicted probabilities on the test set, a threshold of 0.7 was ultimately chosen for classification. The best hyperparameters returned by `gbm_model$bestTune`.

4 Results and Analysis

4.1 Logit Relationship Results

Table 1 and 2 shows how each variable affects customer conversion in terms of log odds in descending order, separated by whether the variable is statistically significant. Statistical significance was determined by the p values of each variable built into the glm function in R. Here is an example on how to interpret the tables: If a customer's email clicks increases by 1, then the log odds of that customer converting increases by 0.1780, ceteris paribus. It is important to note that not all log odds are calculated by the increase of 1, some like loyalty points are higher, per 100. This means that if a customer's loyalty points increases by 100, then the log odds of that customer converting increases by 0.0261, ceteris paribus. A trend that can be seen throughout the two tables shows that engagement metrics like time on site and email clicks are the most effective predictors of customer conversion. Conversely demographic metrics like age or gender are not statistically significant in predicting customer conversion. Even though, as shown in the next section, logistic regression does not perform well in prediction metrics, it is helpful for understanding the relationship between the variables and customer conversion.

CampaignTypeConversion	1.108
EmailClicks	0.1780
PagesPerVisit	0.1508
PreviousPurchases	0.1439
TimeOnSite	0.1069
EmailOpens	0.0874
ConversionRate %	0.0626
ClickThroughRate %	0.0590
LoyaltyPoints Per 100	0.0261
WebsiteVisits	0.0239
AdSpend Per 100	0.0172

Table 1: Statistically significant variables

CampaignChannelPPC	0.1493
CampaignChannelReferral	0.1417
CampaignTypeConsideration	0.06007
CampaignChannelSEO	0.04442
CampaignTypeRetention	0.04288
Income per 10k	0.00927
Age	-0.00101
SocialShares	-0.00184
CampaignChannelSocialMedia	-0.00302
GenderMale	-0.02438

Table 2: Statistically insignificant variables

4.2 Results and Comparison of Logit, RF, SVM, and GBM

Below are the confusion matrices and the performance metrics for the four models after hyperparameter tuning.

	Actual Pos	Actual Neg
Pred Pos	1378	152
Pred Neg	15	37

Table 3: Logit

	Actual Pos	Actual Neg
Pred Pos	1201	54
Pred Neg	201	144

Table 4: RF

	Actual Pos	Actual Neg
Pred Pos	1211	63
Pred Neg	177	132

Table 5: SVM

	Actual Pos	Actual Neg
Pred Pos	1346	62
Pred Neg	42	133

Table 6: GBM

	Logit	RF	SVM	GBM
Balanced Accuracy	0.5925	0.7919	0.7747	0.8259
Sensitivity	0.9892	0.8566	0.8725	0.9697
Specificity	0.1958	0.7272	0.6769	0.6821
F1 Score	0.9429	0.9040	0.9098	0.9628
ROC AUC	0.7766	0.8161	0.8124	0.8699

Figure 4: Final performance metrics

Here are the optimal hyperparameters for the three predictive models:

- RF: `n_estimators = 400`, `max_depths = 10`, `min_sample_split = 5`, `class_weight = balanced`, and `threshold = 0.7`
- SVM: `sigma = 0.0347`, `c = 1`, and `threshold = 0.8432`
- GBM: `n.trees = 200`, `interaction.depth = 1`, `shrinkage = 0.1`, and `threshold = 0.7`

It can be seen through its balanced accuracy and auc that logit performs significantly worse for this task of predicting customer conversion compared to the three other predictive models. This is due to the fact that the dataset is inherently skewed, having over 80 percent of customers convert. This can also be noticed in its extremely low specificity showing that it has a high false positive rate. Ultimately, the purpose of logit is to explain the relationships of the variables to customer conversion via log odds. For predictive accuracy, the other three models are more competitive but a clear winner can still be seen. GBM has a similar number of false positives, but a significantly lower number of false negatives when compared to RF and SVM. This can also be seen through a comparison of their balanced accuracies and AUC. GBM is the only model that achieved a balanced accuracy of over 80 percent, showing that out of the four models, it performs best for the task of predicting customer conversion. RF has the lowest number of false positives but also the highest number of false negatives, showing that it is more conservative compared to the other models. This results in a higher likelihood of missing potential conversions. SVM's performance is moderate, but it provides a valuable comparison benchmark, and its relatively balanced error profile makes it a meaningful alternative. From a business perspective, GBM with a threshold set to 0.7 can provide a suitable balance between conversion rate acquisition and marketing efficiency.

5 Conclusion

The purpose of the research done was to determine the driving factors behind customer conversion and to train a model that can accurately predict whether a customer converts or not. The results gained from the statistical and predictive models suggests that the agency can indeed boost customer conversion rate using metrics collected from the dataset. From the statistical analysis done through logistic regression, it can be seen that the variables that helps with customer conversion are all engagement metrics and that demographic metrics are irrelevant to customer conversion. If the agency can find ways to increase these engagement metrics, the rate of customer conversion can definitely be improved. The agency can also use GBM to help with predicting customer conversion with a pretty good accuracy and to determine what combinations of engagement metrics they would like to increase in customers.

There are also some future improvements that could be made that should be mentioned. First, the dataset itself is very skewed, which begs the question of whether the data is representative of what is actually happening. If not, having representative data would greatly help with the validity of the results. Second, there are very little variables regarding how the agency is attempting to increase customer conversion, with only `adspend`, `campaign type`, and `campaign channel` being related. If provided with more of these types of variables, it could also be possible to determine how to best allocate the agency's resources. Lastly, if some form of resampling was done on the dataset to fix its skewness for each of the predictive models, there could be possible improvements in their predictive abilities.

Contributions statement

Student ID: 39982475 Report: Sections 3.2, 3.3.2, 4, 5. Code: `logit_model.R` and `svm_model.R`

Student ID: 39982483 Report: Sections 1, 3.1, 3.3.1. Code: `exp_data_analysis_plus_vis.py` and `RFmodel.py`

Student ID: 39955478 Report: Sections 2.1-2.5, 3.3.3. Code: `DSF GW Pt1 Preprocess.Rmd` and `gbm_model.R`.

References

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